Loucas Pillaud-Vivien

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Academic experience

2024– **Assistant Professor**, *CERMICS (UMR CNRS 9032), Ecole des Ponts ParisTech – Institut Polytechnique de Paris*, Marne-La-Vallee, France
Member of the Hi! PARIS cluster (Institut Polytechnique de Paris – HEC Paris center in AI and data analytics).

2022–2023 **Courant Instructor - Flatiron research fellow**, *New-York University/Simons Institute*, New-York, United States

2020–2022 **Postdoctoral studies in machine learning at EPFL**, *Supervisor : Nicolas Flammarion, TML*, Lausanne, Switzerland

2017–2020 **Ph.D student in machine learning in the Sierra team-project**, *Supervisors : Francis Bach, Alessandro Rudi INRIA/ENS*, Paris

2016–2017 **Master Specialise at Ecole Nationale des Ponts et Chaussees**, *Politique et Actions Publique pour le Developpement Durable (PAPDD)*, Marne-La-Vallee

2015–2016 **Master 2 at Universite Pierre et Marie Curie (Paris VI)**, *Analyse Numerique et Equation aux Derivees Partielles (ANEDP)*, Paris, Major in PDE (Partial Differential Equation) and Functional analysis

2015– **Eleve-Ingenieur des Ponts, des Eaux et des Forets**

2012–2015 **Student at Ecole Polytechnique**, Palaiseau
Major in applied mathematics and theoretical physics

2009–2012 **Classes preparatoires**, *Lycee Louis-Le-Grand*, Paris
Personal project : Random walks and Brownian motion.

2007–2009 **Lycee**, *Lycee Louis-Le-Grand*, Paris
Baccalaureat in Science with Highest Honors

Teaching Experience

2025 First year research project at ENPC, (Group of 3 students).

2024 Assistant Professor of Statistics and Data Science at ENPC, (25 students during 1 semester).

2023 Research project at Courant Institute, NYU, (Group of 20 students)

2022-2023 Professor at Courant Institute, NYU, *Honors Linear Algebra* (30 elite students during 1 semester, and 250 students for Linear Algebra during 2 semesters.

Supervision

2025– **Leo Dana**, *Ph.D. student*, co-supervised with Francis Bach

2025– **Remi Surat**, *Ph.D. student*, co-supervised with Julien Guyon, CIFRE fellowship with BNP Paribas (Gregoire Loeper)

2027– **Victor Baillet**, *Ph.D. student*, co-supervised with Raphael Berthier

2026 **Imen Ayadi**, *Postdoctoral researcher*

Student Experience

2016, 4 months **Master thesis in PDEs and Probability**, *Cermics*, Marne-la-Vallee, Supervisors : Tony Lelievre and Julien Reygner
Convergence of the empirical measure in a Fleming-Viot type particle system.

2015, 6 months **Master thesis in theoretical physics**, *College de France*, Paris, Supervisors : Antoine Georges and Michel Ferrero
Strongly correlated supraconductivity studied with a Quantum Monte-Carlo algorithm and a Slave-Rotor technique.

2014, 3 months **4G Virtualization and web platform development**, *Orange Sillicon Valley*, San Francisco, USA

2012–2013, 6 months **Teaching internship**, *Lycee Janson-de-Sailly*, Paris
Teaching in a social French program "Cordees de la Reussite" to promote diversity in higher education, especially in classes preparatoires.

Successes

2020 Inria-EPFL International lab Post-Doctoral fellowship "stochastic dynamics in high dimension" (co-chaired with T.Lelievre and N.Flammarion).

Scientific Prize **Le Rivot Prize** 2015 from the **French Academy of Sciences** for the best research internship of the Ecole Polytechnique.

Art Participation in the creation of an artistic experience for Anais Tondeur's art exposition : *LOST IN FATHOMS, Nuuk Island.*

Research Grants

2025– **ANR GALA** (ANR-25-CE23-1585), Principal Investigator. French National Research Agency, approx. 400,000 EUR.

2025– **ANR DysLos** (ANR-25-CE40-6875-01), member of the consortium. French National Research Agency.

2024 **PEPS JCJC** grant, CNRS, 5,000 EUR.

Published Articles

[21] L. Pillaud-Vivien, A. Schertzer
Joint Learning in the Gaussian Single Index Model. *Preprint*, 2025.

[20] L. Dana, F. Bach, L. Pillaud-Vivien
Convergence of Shallow ReLU Networks on Weakly Interacting Data. *Preprint*, 2025.

[19] A. Schertzer, L. Pillaud-Vivien
Stochastic Differential Equations models for Least-Squares Stochastic Gradient Descent. *JMLR*, 2025.

[18] C. Margossian, L. Pillaud-Vivien, L.K. Saul.
Variational inference for uncertainty quantification: an analysis of trade-offs. *JMLR*, 2025.

[17] A. Damian, L. Pillaud-Vivien, J. Lee, J. Bruna
Computational-Statistical Gaps in Gaussian Single-Index Models. *COLT*, 2024.

[16] D. Cai et al.
Batch and match: black-box variational inference with a score-based divergence *ICML*, 2024.

[15] A. Bietti, J. Bruna, L. Pillaud-Vivien.
On Learning Gaussian Multi-index Models with Gradient Flow. *Communication in Pure and Applied Mathematics*, 2024.

[14] JA Varre, M-L Vladarean, L Pillaud-Vivien, N Flammarion.
On the spectral bias of two-layer linear networks. *NeurIPS*, 2023.

[13] J Bruna, L Pillaud-Vivien, A Zweig.
On Single Index Models beyond Gaussian Data. *NeurIPS*, 2023.

[12] L Pillaud-Vivien, F Bach.
Kernelized diffusion maps. *COLT*, 2023.

[11] M Andriushchenko, A Varre, L Pillaud-Vivien, N Flammarion.
SGD with large step sizes learns sparse features. *ICML*, 2023.

[10] E Boursier, L Pillaud-Vivien, N Flammarion.
Gradient flow dynamics of shallow ReLU networks for square loss and orthogonal inputs. *NeurIPS*, 2022.

[9] LP Vivien, J Reygner, N Flammarion.
Label noise (stochastic) gradient descent implicitly solves the Lasso for quadratic parametrisation. *COLT*, 2022.

[8] S Pesme, L Pillaud-Vivien, N Flammarion.
Implicit bias of sgd for diagonal linear networks: a provable benefit of stochasticity. *NeurIPS*, 2021.

[7] V. Cabannes, L. Pillaud-Vivien, F. Bach, A. Rudi.
Overcoming the curse of dimensionality with Laplacian regularization in semi-supervised learning. *NeurIPS*, 2021.

[6] A. Varre, L. Pillaud-Vivien, N. Flammarion.
Last iterate convergence of SGD for Least-Squares in the Interpolation regime. *NeurIPS*, 2021.

[5] L. Pillaud-Vivien.
Learning with Reproducing Kernel Hilbert Spaces: Stochastic Gradient Descent and Laplacian Estimation. *Ph.D. thesis*, 2020.

[4] L. Pillaud-Vivien, F. Bach, T. Lelievre, A. Rudi, G. Stoltz.
Statistical Estimation of the Poincare constant and Application to Sampling Multimodal Distributions. *AISTATS*, 2019.

[3] T. Lelievre, L. Pillaud-Vivien, J. Reygner.
Central Limit Theorem for stationary Fleming-Viot particle systems in finite spaces. *ALEA*, 2018.

[2] L. Pillaud-Vivien, A. Rudi, F. Bach.
Statistical Optimality of Stochastic Gradient Descent on Hard Learning Problems through Multiple Passes. *NeurIPS*, 2018.

[1] L. Pillaud-Vivien, A. Rudi, F. Bach.
Exponential convergence of testing error for stochastic gradient methods. *COLT*, 2018.

Academic Service

Area Chair NeurIPS 2026.

Reviewer *Journals:* Annals of Applied Probability, Foundations of Computational Mathematics, Annals of Statistics, JMLR, IEEE Transactions on Information Theory, SIAM Journal on Applied Dynamical Systems, SIAM Journal on Mathematics of Data Science, Applied and Computational Harmonic Analysis, IoP Science-Machine Learning: Science and Technology.

Conferences: NeurIPS, ICML, ICLR, ALT.

Languages

French native speaker

English fluent

Spanish, German basics

Computer skills

Text Microsoft Office, LateX

Coding Python, Php, HTML

Interests

Cultural activities Contemporary Art and Theater

Sports Surf, Tennis and Football in competition.